

Artificial Neural Networks

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Course outline

❑ Introduction to artificial intelligence (1.5h)

❑ Understanding Neural Networks (3.5h)

- From Perceptron to Deep Learning
- Activation Functions
- Types of tasks

❑ How Neural Networks Learn (4h)

- Training cycle
- Gradient Descent
- Backpropagation
- Optimizers

Lab 01. PyTorch
Fundamentals (1h)

Lab 02. Binary
Classification (3h)



Course outline

□ Model Evaluation & Improvement (3h)

- Learning, Evaluation, Testing
- Underfitting and Overfitting
- Hyperparameter tuning
- Regularization Methods

Lab 03. Regression
Regularisation (2h)

Lab 04. Regression +
Hyperparameter
Tuning (2h)

Lab 05. Multiclass
classification –
complete case (3h)





Isaac Asimov - Robot series I (ill. Billy Nunez)

Introduction to Artificial Intelligence

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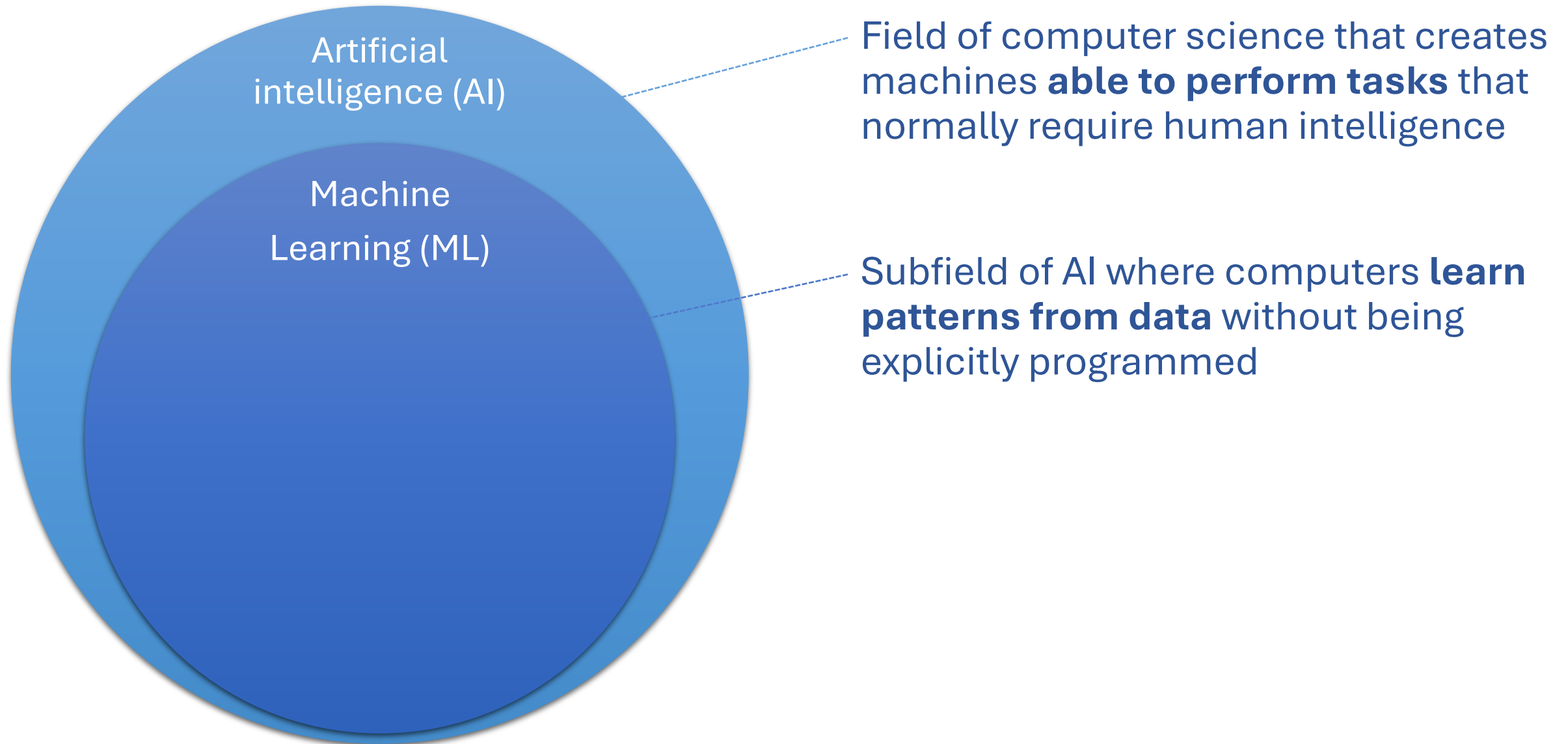
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What is artificial intelligence?

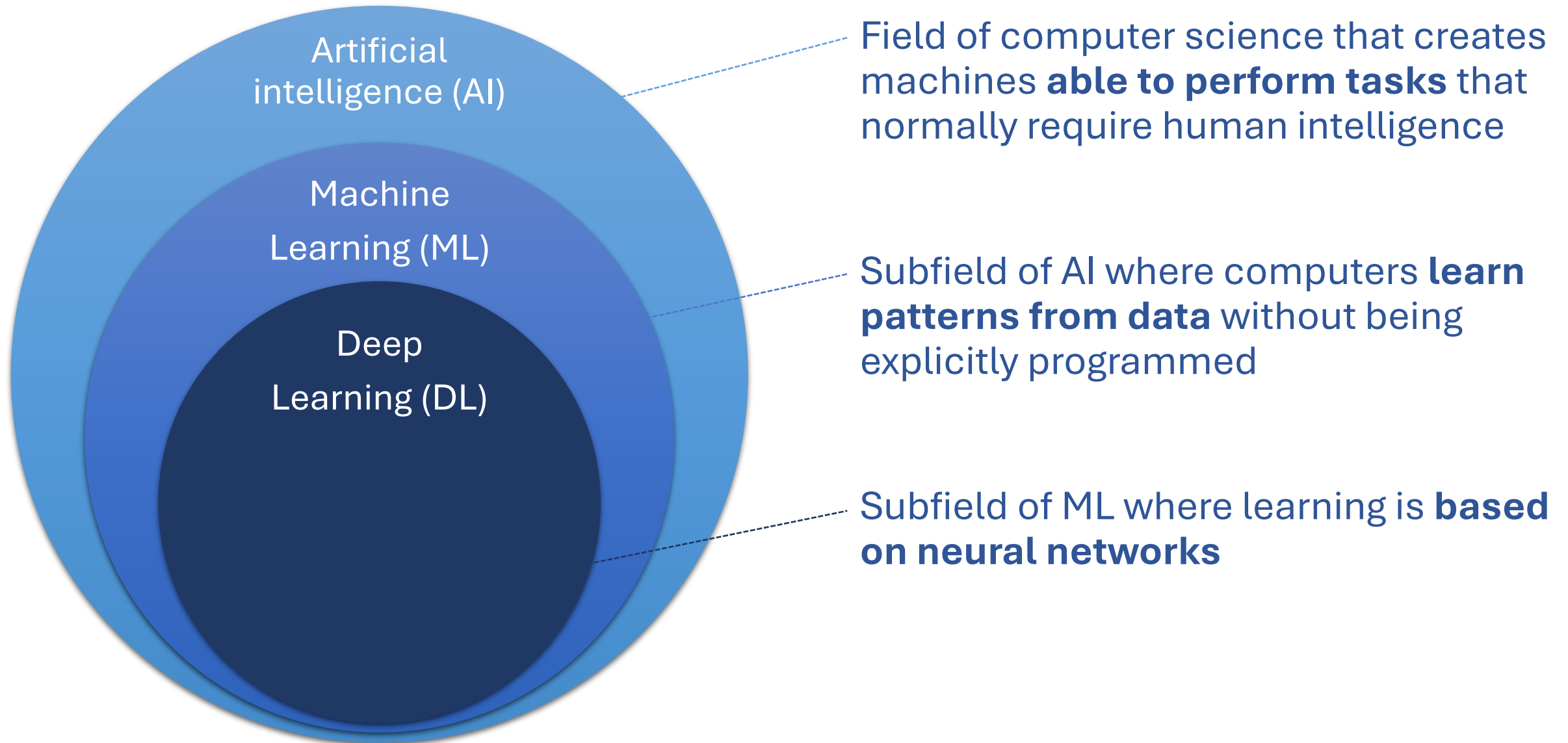
Artificial
intelligence (AI)

Field of computer science that creates machines **able to perform tasks** that normally require human intelligence

What is artificial intelligence?



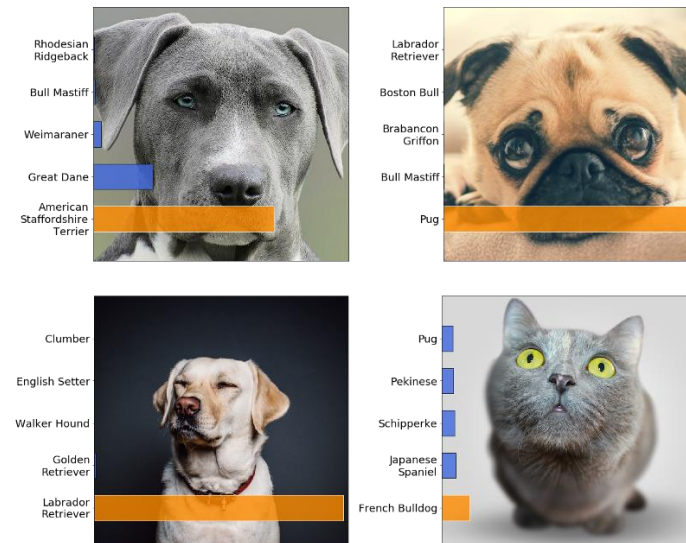
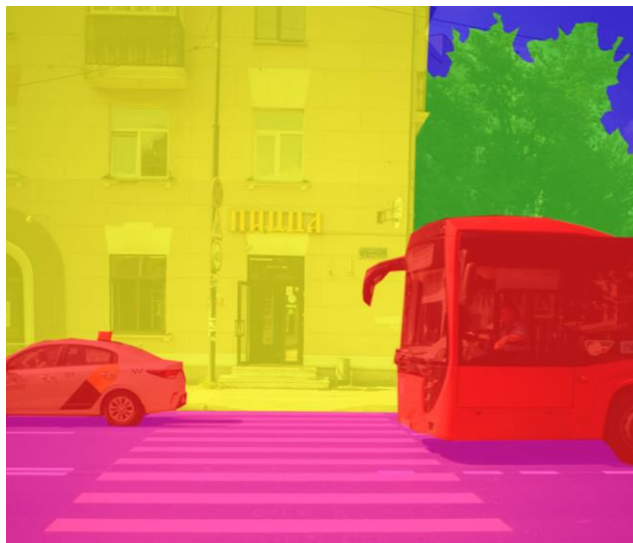
What is artificial intelligence?



Examples of Deep Learning applications

❑ Computer Vision

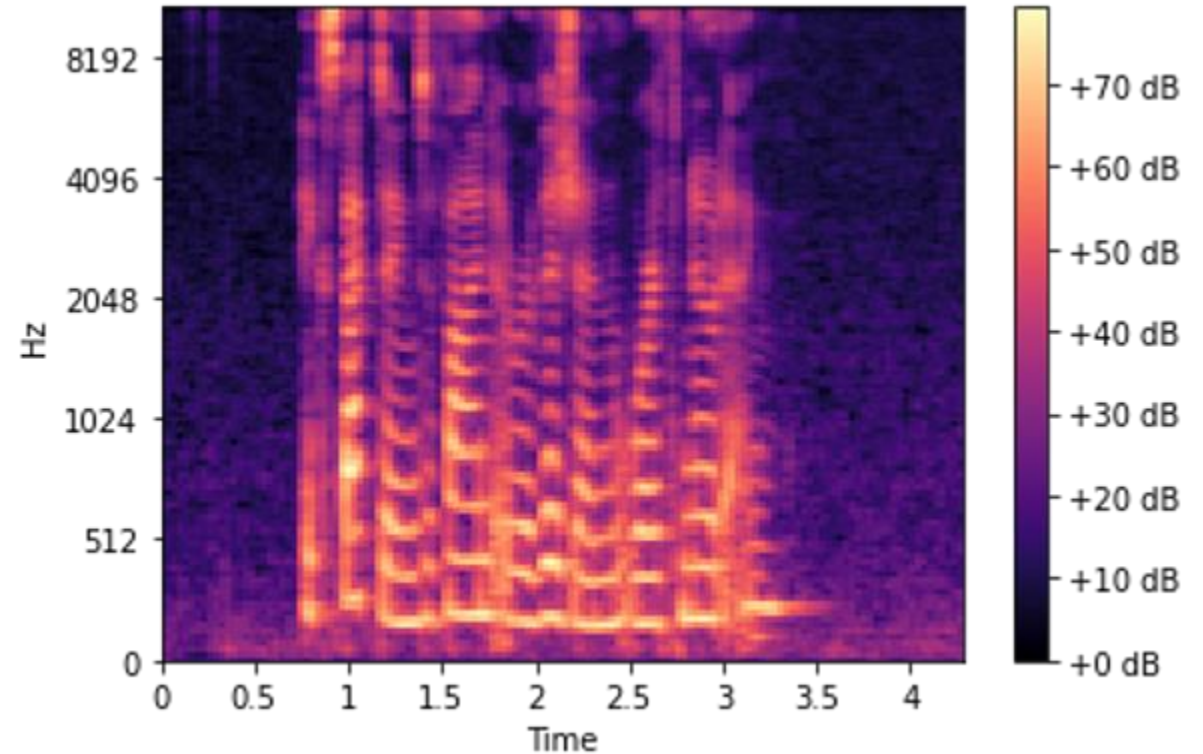
- **Image classification** (e.g., medical imaging: tumour detection)
- **Object detection** (e.g., self-driving cars)
- **Image segmentation** (e.g., satellite mapping, agriculture)
- **Face recognition** (e.g., security, smartphones)



Examples of Deep Learning applications

□ Speech & Audio

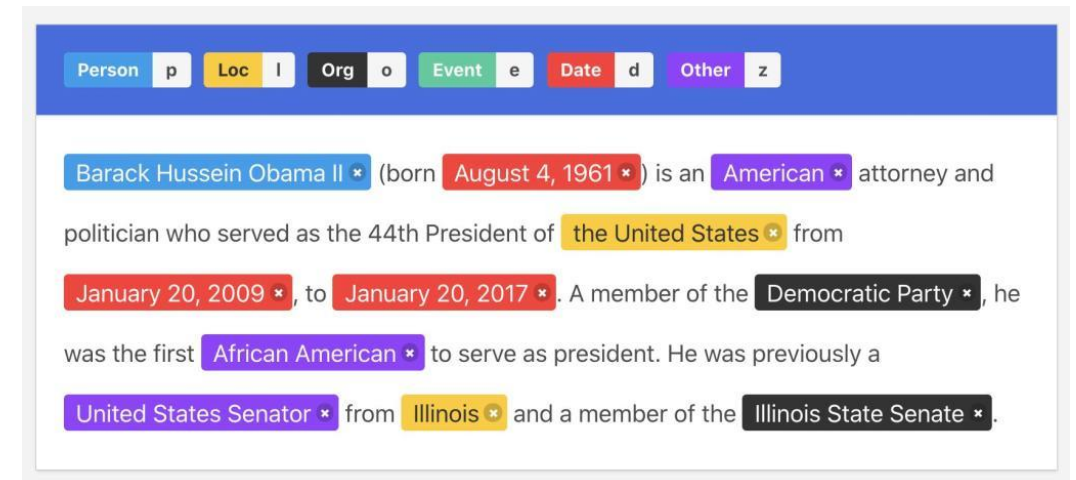
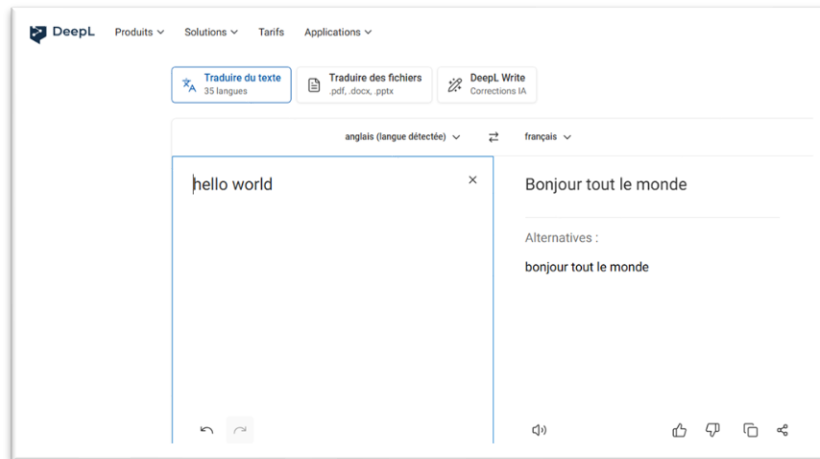
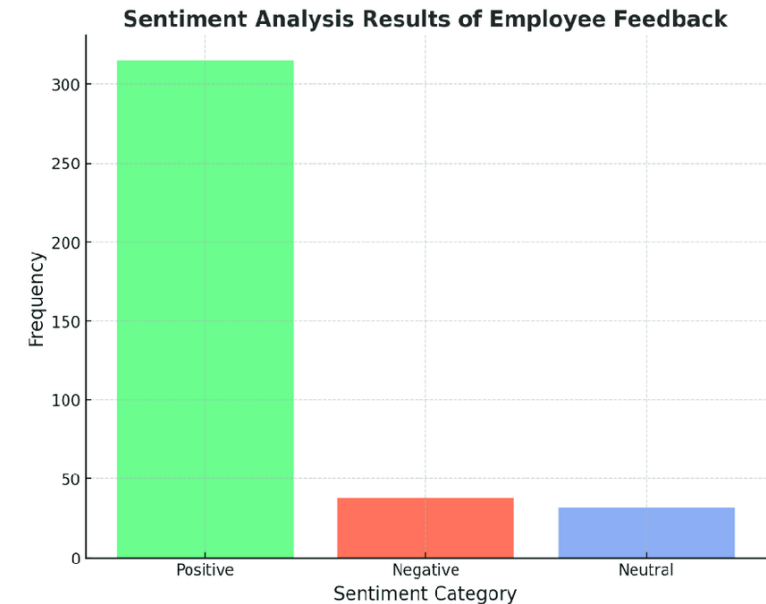
- **Automatic Speech Recognition (ASR)**
(e.g., Siri, Alexa, Whisper)
- **Speaker identification**
- **Music generation & audio synthesis**



Examples of Deep Learning applications

❑ Natural Language Processing (NLP)

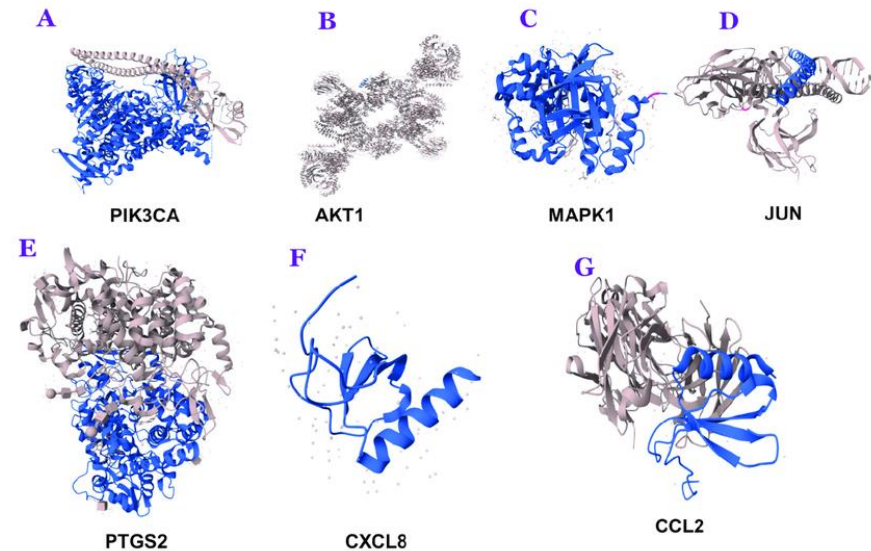
- Text translation (e.g., Google Translate)
- Sentiment analysis (e.g., customer feedback monitoring)
- Chatbots & virtual assistants (e.g., ChatGPT)
- Text summarization and question answering



Examples of Deep Learning applications

❑ Other Domains

- **Recommendation systems** (Netflix, Amazon, Spotify)
- **Fraud detection** (banking, insurance)
- **Drug discovery & genomics** (bioinformatics)
- **Game playing** (AlphaGo, reinforcement learning agents)



Examples of ML methods that are not Deep Learning

- ❑ Linear Regression
- ❑ Logistic Regression
- ❑ Decision Trees
- ❑ Random Forests
- ❑ Gradient Boosted Trees (XGBoost, LightGBM, CatBoost)
- ❑ Support Vector Machines (SVMs)
- ❑ Naive Bayes Classifier
- ❑ k-Nearest Neighbors (kNN)
- ❑ Clustering algorithms (K-means, DBSCAN, Hierarchical clustering)
- ❑ Dimensionality reduction methods (PCA, t-SNE, UMAP)

Examples of AI methods that are not ML

❑ **Rule-based systems / Expert systems**

- e.g., medical diagnosis systems from the 1980s (MYCIN)
- Hand-crafted “if-then” rules

❑ **Search algorithms**

- Minimax search (chess, checkers, tic-tac-toe)
- A* algorithm (robot pathfinding, GPS navigation)

❑ **Classical robotics (before DL perception)**

- Hard-coded rules for robot navigation and control

The machine learning workflow

1 Problem definition & goal setting



The machine learning workflow

- 1 Problem definition & goal setting
- 2 Data collection & preparation



The machine learning workflow

- 1 Problem definition & goal setting
- 2 Data collection & preparation
- 3 Model selection & training



The machine learning workflow

- 1 Problem definition & goal setting
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- 4 Model evaluation & improvement



The machine learning workflow

- 1 Problem definition & goal setting
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- 4 Model evaluation & improvement
- 5 Model deployment & usage



The machine learning workflow

- 1 Problem definition & goal setting
- 2 Data collection & preparation
- 3 Model selection & training
- 4 Model evaluation & improvement
- 5 Model deployment & usage
- 6 Model Monitoring & Maintenance



1 Problem definition & goal setting

- ❑ What do we want to predict?
- ❑ What data is available?
- ❑ What are the success metrics?



2 Data collection & preparation

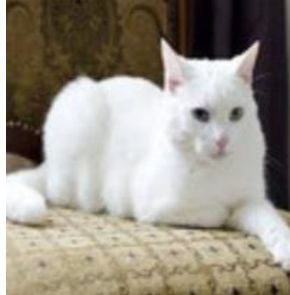
- ❑ The model will learn data in order to make accurate **predictions**
- ❑ Data can be **labelled**: categories are defined by humans.
 - X features, **1 target**
- ❑ Data can be **un-labelled**. The model extracts the features and defines the categories by itself
 - X features, **no target**



The machine learning workflow

2 Data collection & preparation

- Example of labelled data:
 - Categorical target

Input data	Label (target)
	<i>"Cat"</i>
	<i>"Bird"</i>
	<i>"Cat"</i>



The machine learning workflow

2 Data collection & preparation

- ❑ Example of labelled data:
 - Numerical target

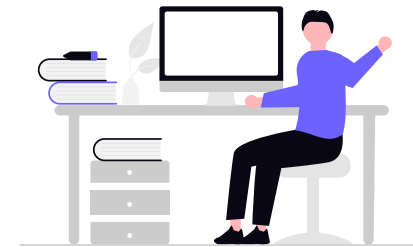


MedInc	House Age	Rooms	Bedrms	Population	Occup	target
4.5375	34.0	5.381944	0.968750	1011.0	3.510417	1.890
3.8792	22.0	4.715254	1.145763	810.0	2.745763	2.182
4.9779	23.0	4.858012	1.010142	1309.0	2.655172	2.246

The machine learning workflow

2 Data collection & preparation

- ❑ Example of labelled data:
 - Categorical target

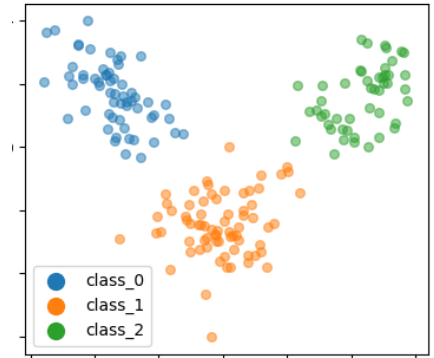


alcohol	Malic acid	ash	Magnesium	Total phenols	Color Intensity	proline	target
12.29	2.83	2.22	88.0	2.45	2.15	290.0	1
13.58	2.58	2.69	105.0	1.55	8.66	750.0	2
11.65	1.67	2.62	88.0	1.92	2.60	562.0	1

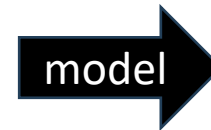
The machine learning workflow

2 Data collection & preparation

- ❑ Example of un-labelled data:



alcohol	Malic acid	ash	Magnesium	Total phenols	Color Intensity	proline
12.29	2.83	2.22	88.0	2.45	2.15	290.0
13.58	2.58	2.69	105.0	1.55	8.66	750.0
11.65	1.67	2.62	88.0	1.92	2.60	562.0



Target
1
2
1

2

Data collection & preparation

- ❑ Basic rules for dataset preparation
 - Data must be balanced between categories
 - Data must be shuffled
 - Data must not have duplicates



2

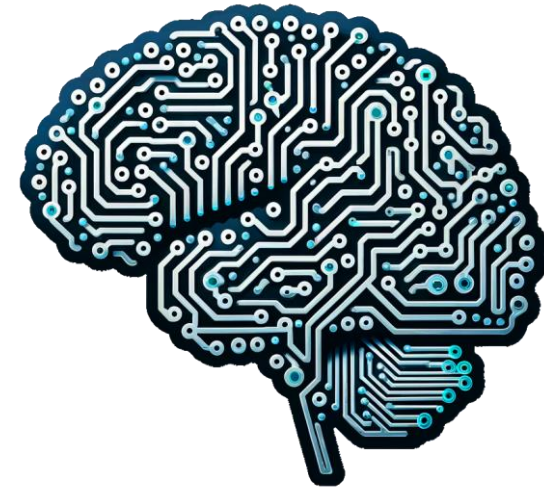
Data collection & preparation

- ❑ Before training, data is split in several subsets:
 - **Training set**, to train the model
 - **Validation set**, to monitor training
 - **Test set**, to evaluate model **after** training



3 Model selection & training

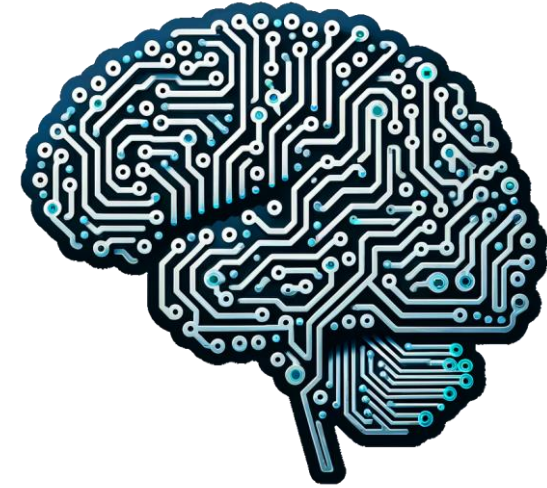
- ❑ Supervised learning
- ❑ Unsupervised learning
- ❑ Semi-supervised learning
- ❑ Reinforcement learning



3 Model selection & training

□ Supervised learning

- Model learns from data labeled by humans
- Model predicts labels for new data
- Examples: Regression, Classification



- High accuracy when labeled data is good
- Unlimited applications

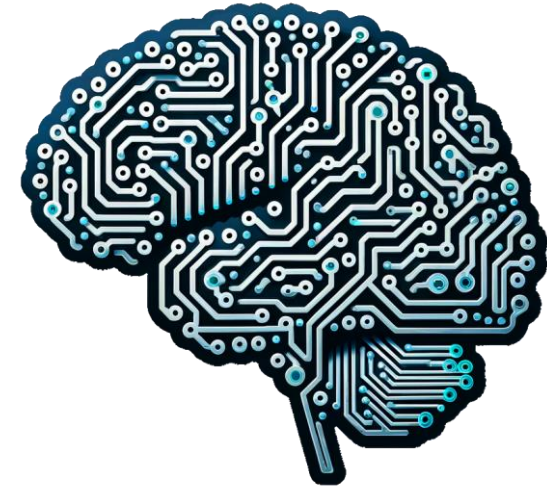


- Dataset labeling by hand is slow and expensive procedure
- Need great quantity of examples

3 Model selection & training

□ Unsupervised learning

- Model finds patterns or structure in unlabeled data
- Model make groups with data, compress data
- Examples: Clustering, Dimensionality Reduction



- No labels required, time, money saved

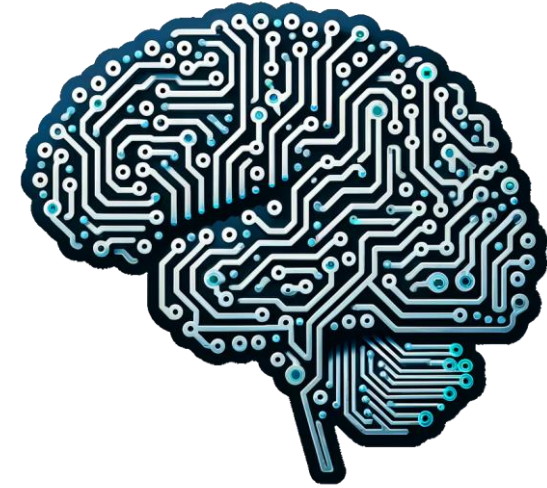


- Hard to evaluate
- Less predictable
- Limited applications

3 Model selection & training

❑ Semi-supervised learning

- Mix of few labeled and many unlabeled examples
- Model learns from labelled data to predict pseudo-labels. Then model is trained again with labelled data + high confidence pseudo-labels



- ❑ When labelled data is difficult to obtain



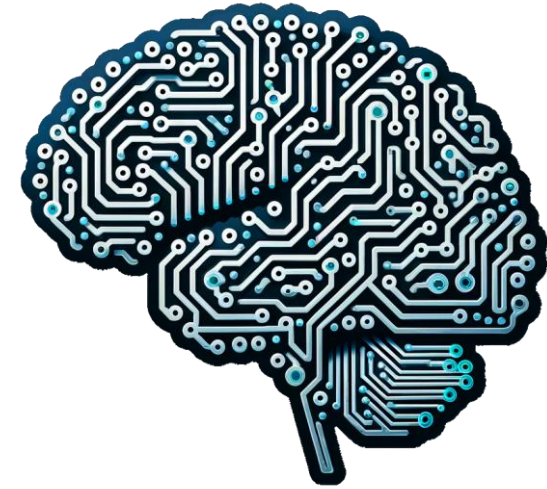
- ❑ Unstable

3

Model selection & training

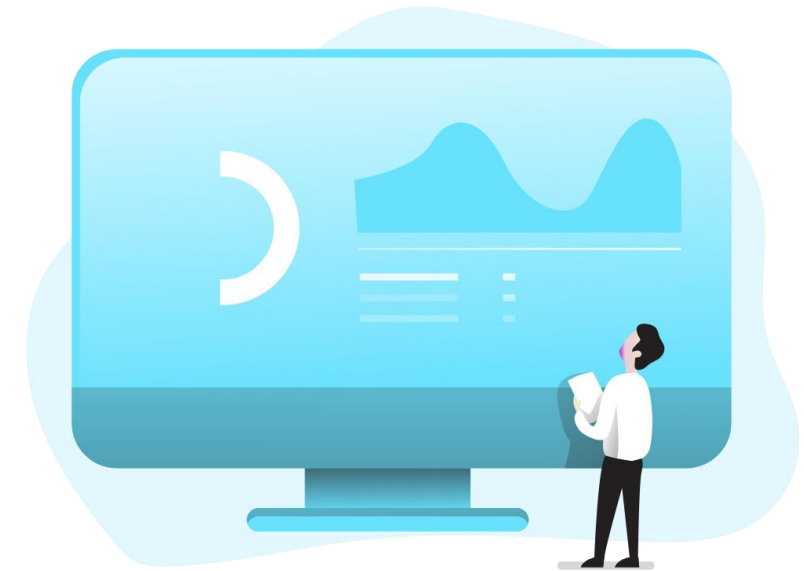
❑ Reinforcement learning

- Model learns by interacting with an environment and receiving feedback
- Model tries to maximize reward over time
- Examples: Robotics, Games



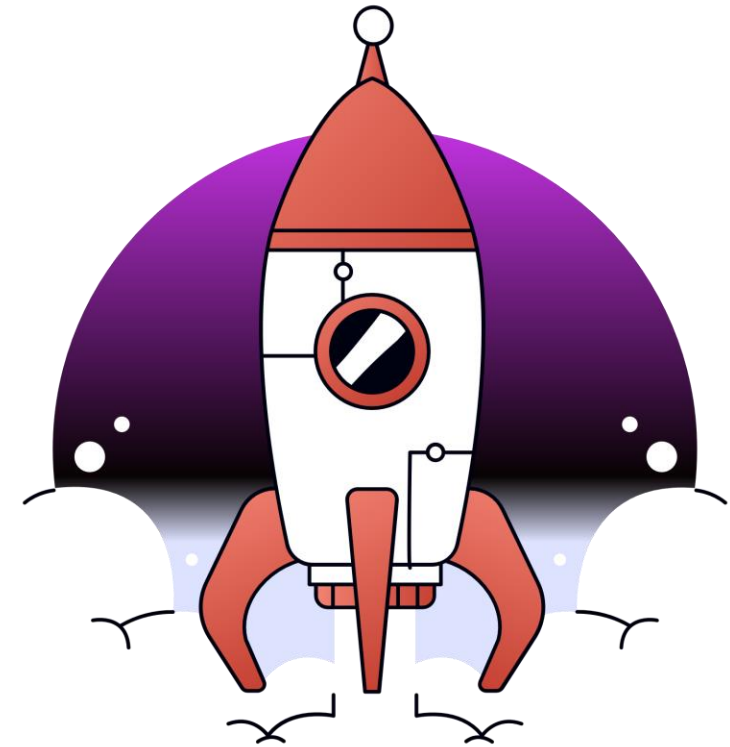
4 Model evaluation & improvement

- ❑ To ensure model is performing well, it **must be evaluated on unseen data**
- ❑ Evaluation requires:
 - A Test dataset
 - One or several evaluation metrics
- ❑ Test and Train dataset **must have the same distribution**



5 Model deployment & usage

- ❑ Deployment is the process of integrating a trained model into a real-world application (apps, APIs) to make predictions on new data
- ❑ A model can be deployed locally (**on premise**) or **on cloud**
- ❑ Usage of a model to make predictions is called **inference**



The machine learning workflow

6 Model Monitoring & Maintenance

- ❑ Once a model is deployed, it is not over...
- ❑ Data can evolve (Data drift) and the model can become less accurate
- ❑ New data need to be acquired and model needs to be retrained
- ❑ Automation of data pipelines and model deployment is possible

MLOPS Cycle

